A. Segmentation: Otsu Thresholding

For the segmentation, I used **Otsu's method**. Otsu is a way to find the threshold automatically, instead of me guessing a number.

First, it counts how many pixels there are for each brightness level from 0 to 255, which is the histogram. Then I try every possible threshold one by one. Each threshold splits the pixels into two groups: the dark group and the bright group. For each one, it check how different the two groups are, by looking at how far apart their average brightness is, and also how big each group is.

The threshold that makes the two groups the most different from each other is the winner. That's the one Otsu picks. After that, every pixel brighter than the threshold becomes white (255), and every pixel that's not becomes black (0). That's the raw segmentation.

Because of that, the bright parts of the tissue (the background, the lumen, the cytoplasm) turn white, and the dark parts (the core and the dense tissue) turn black. This is important for the next part, because erosion and dilation work on the white parts.

B. Morphological Processing

For this, I used the **3×3 cross shaped kernel**. The kernel is like a small plus sign that moves over every pixel of the image, and it looks at the pixel and its 4 neighbors (up, down, left, right) to decide the new value of that pixel. I used the cross because the other kernels yield different results I don’t really prefer and the cross one works best for me.

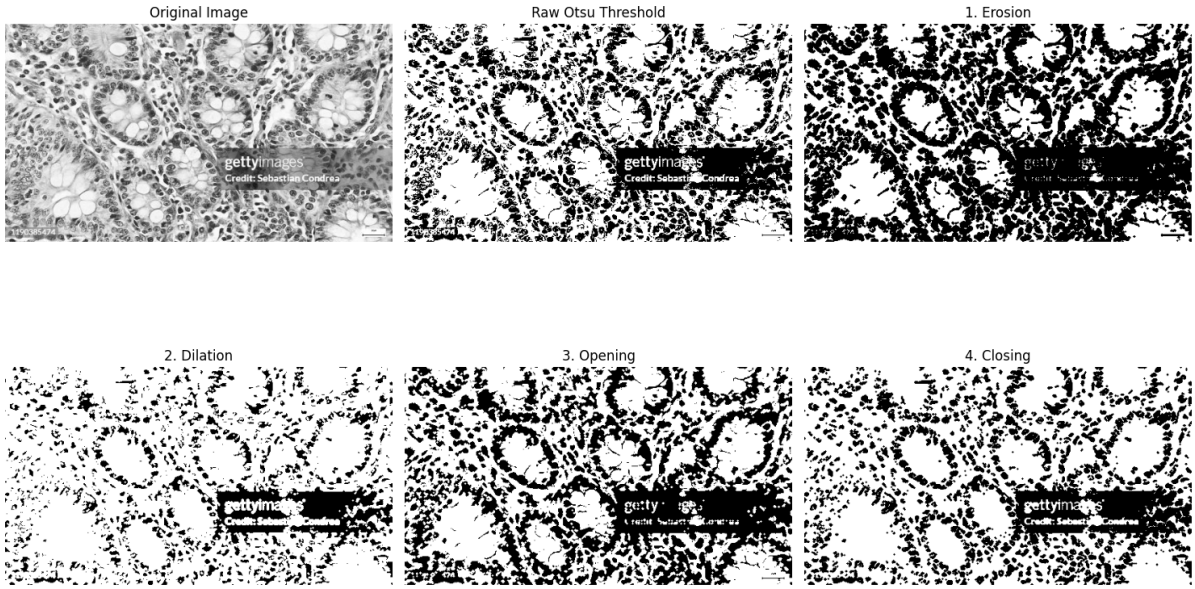
- **Erosion**: a pixel stays white only if all its neighbors under the kernel are white too. If any of them is black, the pixel becomes black. So the white areas shrink, and the small white specks disappear.
- **Dilation**: the opposite. A pixel becomes white if any of its neighbors is white. So the white areas grow, and the small black holes get filled.
- **Opening**: erosion first, then dilation. The erosion removes the small white specks, then the dilation brings the bigger white areas back to about their original size. So it cleans the small white noise.
- **Closing**: dilation first, then erosion. The dilation fills the small black holes and gaps, then the erosion brings the edges back. So it cleans the small black noise inside the white areas.

Erosion and dilation on their own change the image a lot, but opening and closing are more gentle, because the second step undoes most of the size change of the first step. Opening can only remove white pixels, and closing can only add them.

C. Results

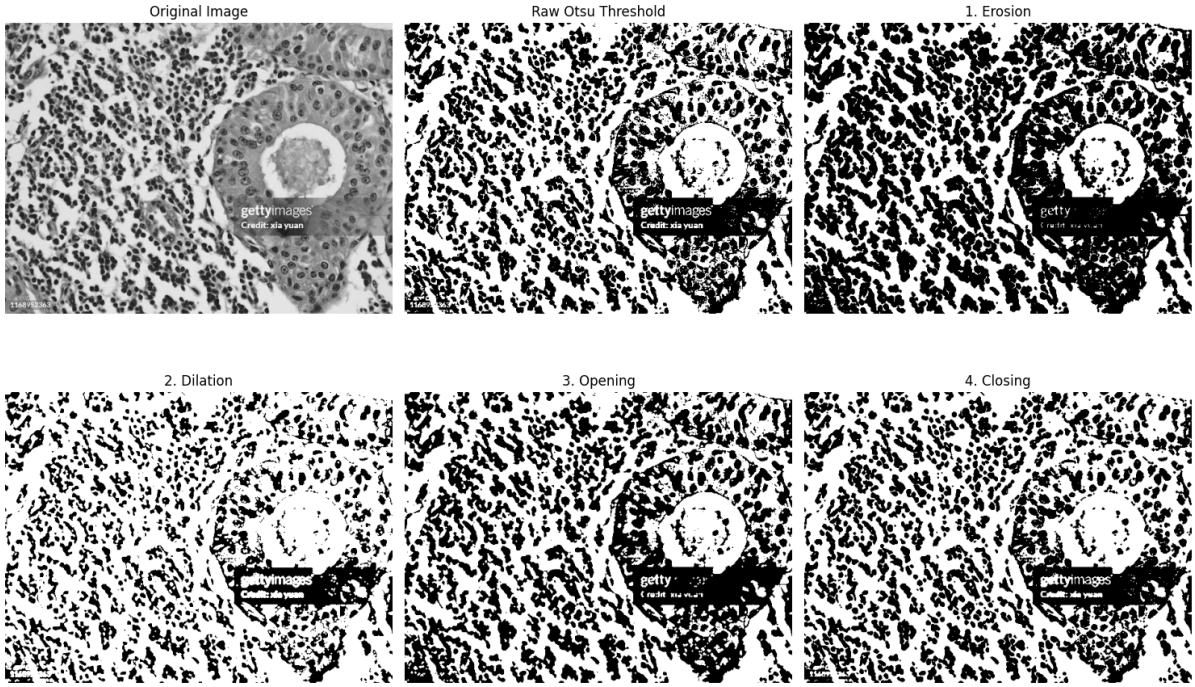
Since white is the bright part, it can look like erosion and dilation are switched at first, but they aren't. It's just that the dark core are black, so erosion makes the black parts look bigger and dilation makes them look smaller.

1. Pancreas (Otsu threshold is 163)



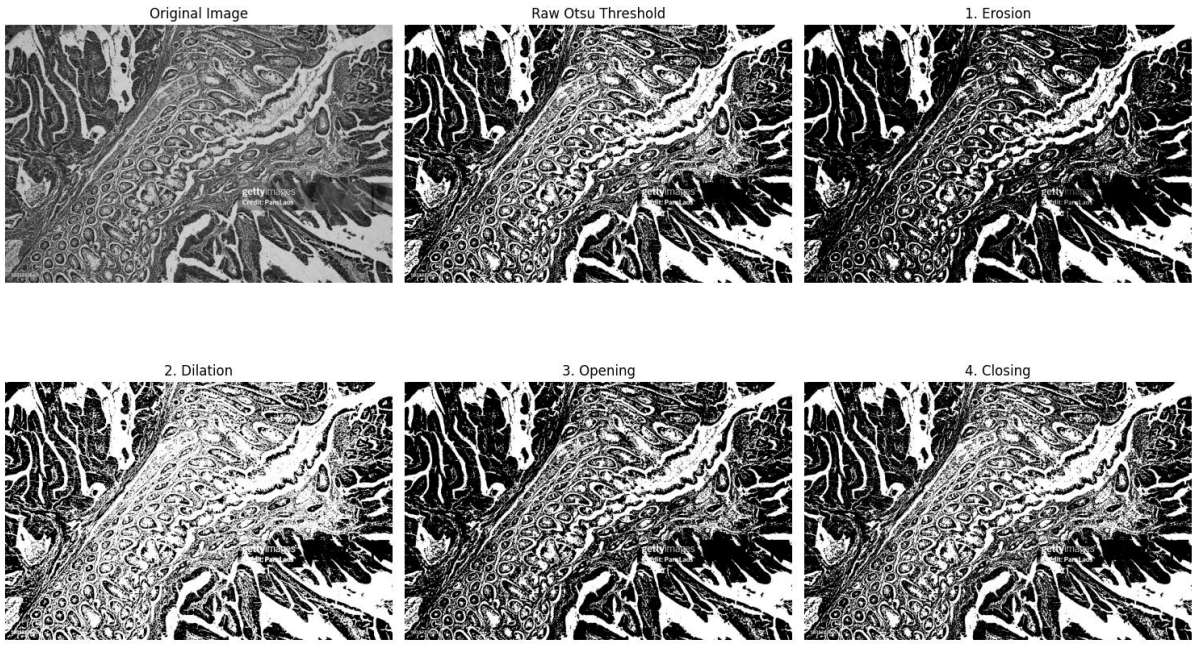
The gland rings and the bright hollow parts are clear in the raw result, but there's still some speckle around the core. Erosion makes the dark borders thicker, and dilation turns most of the image white, where the small core almost disappear. Opening removes the small white specks, and closing gives the smoothest look of all, although some small dark core get filled in.

2. Cancer (Otsu threshold is 143)



This one is full of small dark core, so they show up as black blobs on a white background. With erosion, the black blobs get bigger and a lot of them stick together into clumps. With dilation, the blobs shrink and many of them break apart. Opening makes the gland area on the right look more solid, since the small white specks inside it are removed. Closing removes some of the small black bits in the bright area. For this image, the raw, opening, and closing results are the most useful, because the core stay separate from each other

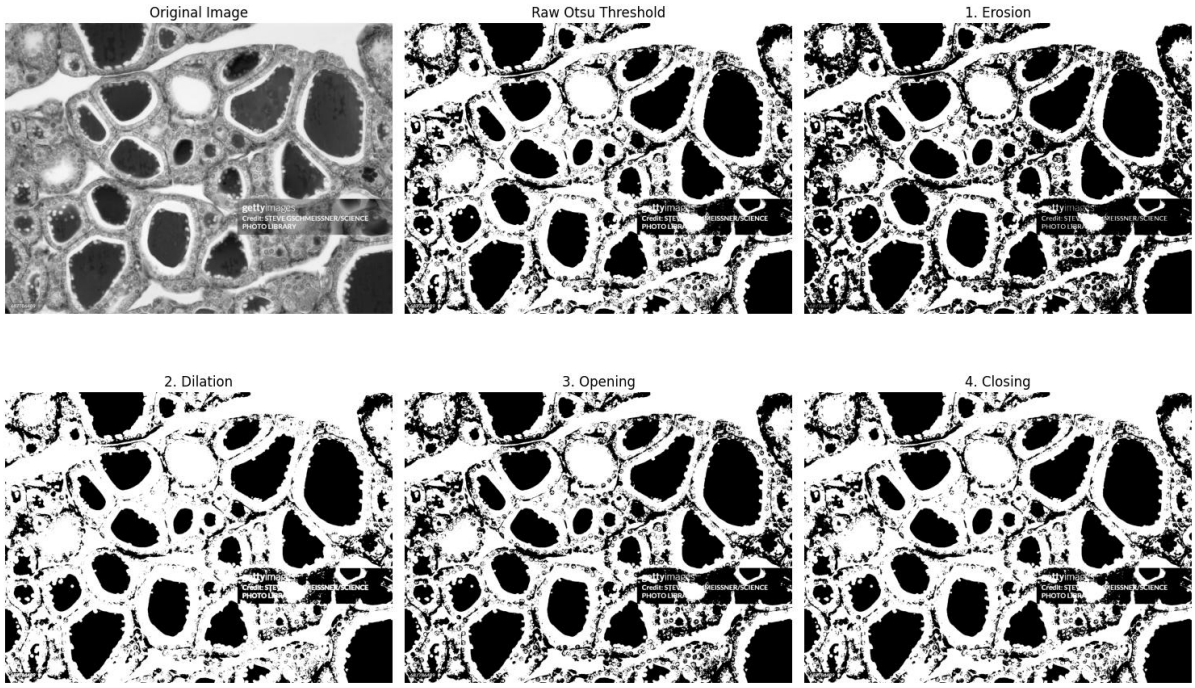
3. Tumor (Otsu threshold is 125)



This image is zoomed out, so it has a lot of small gland shapes. Otsu separates the dark tissue from the bright gaps quite well, and the ring patterns of the glands are still visible in the

middle. Because the details are reallllly fine here, erosion changes it the most. The thin white rings almost disappear and the middle becomes mostly black. Dilation does the opposite and fills the rings. Opening and closing look almost the same as the raw one, since the small kernel only touches the small specks.

4. Thyroid (Otsu threshold is 146)



This is the cleanest result of the four. The big cells with a large core turn into black shapes, and the thin walls around them stay white, because the contrast between them is strong. Since the shapes are big, the morphology doesn't change much here. The differences are mostly on the walls and around the small cell center. Erosion makes the walls thinner, dilation makes them thicker and cleaner, and opening and closing are almost the same as the raw one, with just a bit small specks removed.

D. Conclusion

In the end, Otsu worked well for the images where the dark and bright parts are clearly different, like the thyroid and cancer images. It worked less well on the tumor image, which has a lot of fine detail. How much the morphology changes the image depends on how big the shapes are compared to the kernel. For small and thin details, the change is big, but for big shapes, it only changes the edges. Opening and closing gives a gentler improvement compared to the erosion and dilation, although the difference compared with the otsu isn't quite high (it's still a bit better though). So overall, **I think otsu+opening or closing is the best.** While otsu alone is alright, opening or closing gives it better tweaks in the end.